

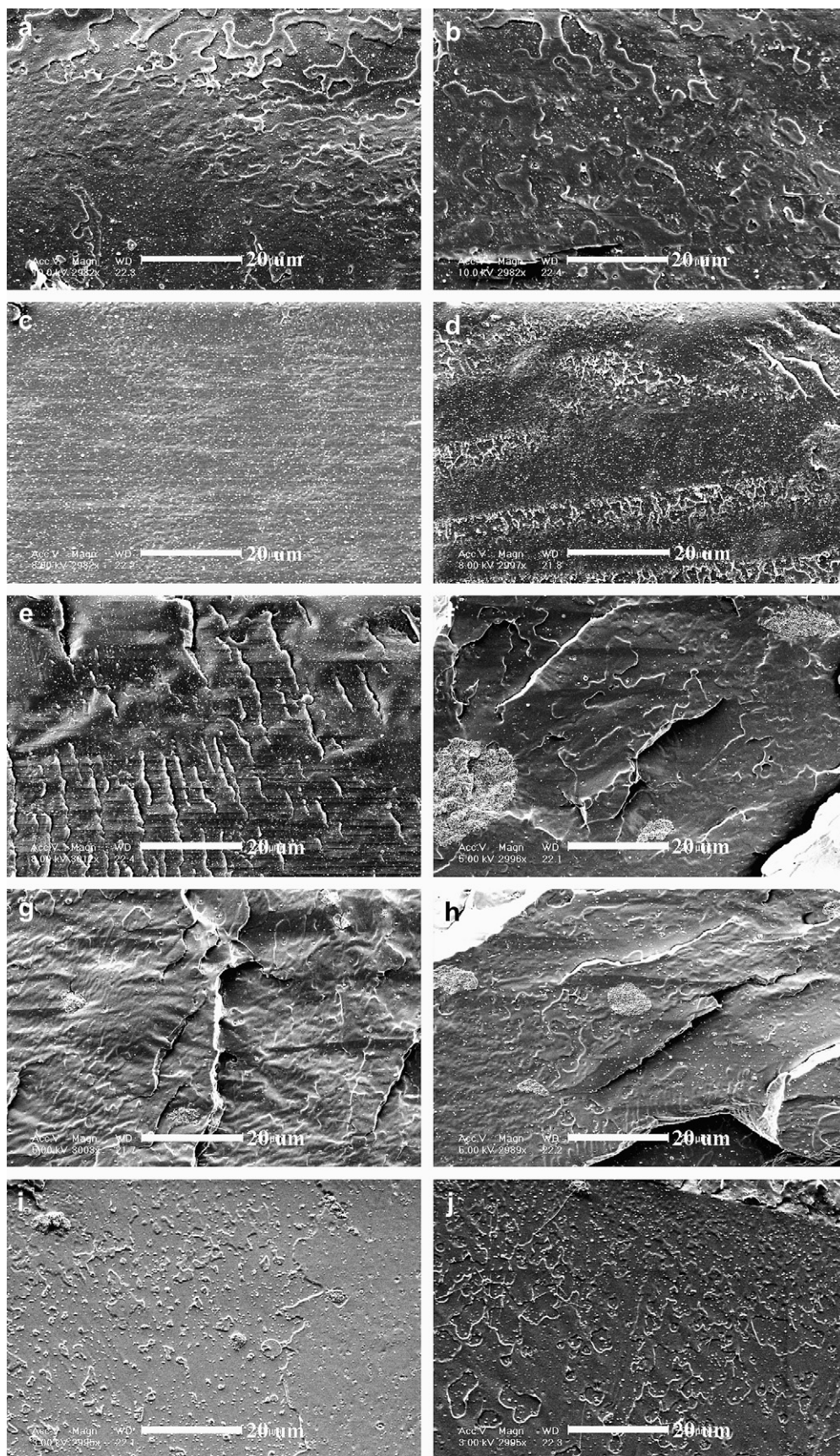


Fig. 8. SEM images of PLLA composites prepared via melt compounding under different screw rotation speed (a–e) PLLA/MWNT (f–j) PLLA/MWNT-g-POSS. The content of the MWNTs was 1% by weight. (a, f) 100 rpm, (b, g) 500 rpm, (c, h) 1000 rpm, (d, i) 1500 rpm, (e, j) 2000 rpm.

and the resulting SEM images are demonstrated in Fig. 7. The bright regions in the SEM images are attributed to MWNTs as a result of their high electrical conductivity. It is unambiguous from Fig. 7 that the MWNT-g-POSS (b) is homogeneously dispersed in the PLLA matrix without any aggregation while obvious aggregation is clearly observed in the PLLA/MWNT (a) composite.

Fig. 8 reveals the dispersion of MWNTs in the PLLA matrix during the melt compounding in which the different screw rotation speeds, namely from 100 to 2000 rpm, were selected. For the polymer composites with the functionalized additives, such as carbon nanotube and clay, the mixing time and shear force are the key factor controlling the morphological structure [45]. Applying a shear force should break up the primary particles and increase the uniformity. The homogeneous dispersion of additives in the polymer matrix is one of the most important requisites for the mechanical strength reinforcement because inhomogeneities normally lead to structural defects in the composite materials.

The SEM micrograph in Fig. 8a–e reveals a homogeneous dispersion of MWNTs throughout the PLLA/MWNT-g-POSS composites without aggregation or tactoid formation, which is indicative of a fully exfoliated structure of the PLLA/MWNT-g-POSS composite when the compounding was carried out at any rotation speed. In contrast, as shown in Fig. 8f–j, the dispersion of MWNTs is significantly depended on the screw rotation speed of the extruder. Aggregates are clearly observed in the PLLA/MWNT composites under low shearing, which means rotation speed is lower than 1500 rpm. At the same MWNT loading, composites with poor nanotube dispersion have discrete nanotube-rich domains rather than a nanotube network, such that the polymer chains flow independent of the nanotubes, and their motion is similar to that of pure PLLA. The dispersion of PLLA/MWNT becomes better when the rotation speed of extruder is higher than 1500 rpm.

DSC was employed to evaluate the effect of MWNTs on the phase transition behavior of PLLA. The glass transition temperature (T_g) of the hybrid composites is associated with a cooperative motion of long-chain segments, which may be hindered by the MWNT [46]. Therefore, as expected, both PLLA/MWNT and PLLA/MWNT-g-POSS composites recorded higher glass transition temperatures than pure PLLA as shown in Fig. 9 and Table 1. It was also found that the enhancement in T_g for PLLA/MWNT-g-POSS composite was more significant than that for PLLA/MWNT composite. It may be suggested that the higher enhancement in T_g for PLLA/MWNT-g-POSS composites is due to the strong interaction between the alkyl chains in POSS and the PLLA chains, and is able to hinder the motion of the polymer chains. Fig. 9 also shows that the melting temperature (T_m) of PLLA/MWNT-g-POSS composite decreased comparing with the pure PLLA. The marked decrease in T_m of PLLA/MWNT-g-POSS composite is probably the result of the CNTs nucleating and speeding up polymer crystallization in the polymer/CNT composites. Resultant imperfect polymer crystals in the composites will melt at a lower temperature than the control polymer.

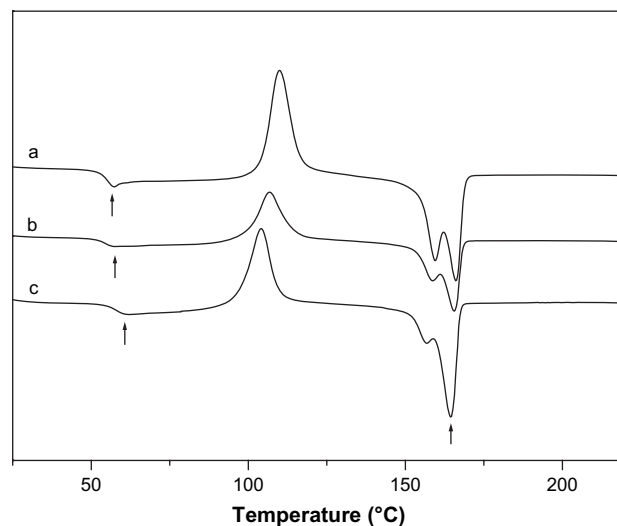


Fig. 9. DSC heating thermograms of (a) PLLA, (b) PLLA/MWNT and (c) PLLA/MWNT-g-POSS composites processing at 190 °C for 4 min under 1000 rpm. The content of the MWNTs was 1% by weight.

The Halpin–Tsai equation has been used successfully to predict the modulus of polymer/carbon nanotube composites [47,48]. Considering the random distribution of carbon nanotubes in the polymer matrix, the modified Halpin–Tsai equation is written as [49,50]:

$$E_C = \left[\frac{3}{8} \frac{1 + 2(l_{\text{CNT}}/D_{\text{CNT}})\eta_L V_{\text{CNT}}}{1 - \eta_L V_{\text{CNT}}} + \frac{5}{8} \frac{1 + 2\eta_T V_{\text{CNT}}}{1 - \eta_T V_{\text{CNT}}} \right] E_P \quad (1)$$

$$\eta_L = \frac{E_{\text{CNT}}/E_P - 1}{E_{\text{CNT}}/E_P + 2l_{\text{CNT}}/D_{\text{CNT}}} \quad (2)$$

$$\eta_T = \frac{E_{\text{CNT}}/E_P - 1}{E_{\text{CNT}}/E_P + 2} \quad (3)$$

where E_C , E_{CNT} , and E_P are the tensile modulus of the composite, nanotube, and polymer matrix, respectively. l_{CNT} , D_{CNT} , and V_{CNT} are the length, diameter, and volume fraction of the nanotubes in the composite, respectively. Aspect ratio ($l_{\text{CNT}}/D_{\text{CNT}}$) of the reinforcing nanotubes can be estimated if E_C , E_{CNT} , E_P , and V_{CNT} are known.

The modulus of the PLLA/MWNT-g-POSS composite (E_C) is plotted as a function of the MWNT-g-POSS loading in Fig. 10, setting the density of MWNT to be $\rho = 2.1 \text{ g/cm}^3$. The average length, l_{CNT} , and diameter of the MWNT, D_{CNT} , were taken to be $l_{\text{CNT}} = 2.0 \text{ }\mu\text{m}$ and $D_{\text{CNT}} = 20 \text{ nm}$. The value of E_{CNT} was assumed to be 1280 GPa.

Table 1

Glass transition temperature (T_g) and melt temperature (T_m) of PLLA, PLLA/MWNT, and PLLA/MWNT-g-POSS composites

Samples	T_g (°C)	T_m (°C)
PLLA	57.2	166.1
PLLA/MWNT	57.8	165.7
PLLA/MWNT-g-POSS	60.1	164.2

The content of the MWNTs is 1% by weight and the rotation speed is 1000 rpm.